

DBMS Normalization - Class Notes [messy but mine]

1NF: no repeating groups. each cell atomic. PK must exist.

Students[id, name, phones] BAD if phones = '98200, 98111' in one cell.

2NF: 1NF + no partial dependency. only matters with COMPOSITE pk.

Enroll[sid, cid, sname, grade]: sname depends only on sid => partial!

fix: Students[sid, sname] + Enroll[sid, cid, grade].

3NF: 2NF + no transitive dependency. non-key must not depend on non-key.

Emp[id, dept, dept_city]: dept_city depends on dept, not id => transitive!

fix: Emp[id, dept] + Dept[dept, dept_city].

Students table example: Students[sid, name, hostel, warden].

warden depends on hostel, hostel depends on sid => NOT 3NF.

fix: Students[sid, name, hostel] + Hostels[hostel, warden].

exam tip: they always ask to convert Enroll or Emp tables. watch for partial vs transitive - partial needs composite key, transitive doesn't.